

Architectural Design and Optimization of Project V.E.S.P.A.: A Real-Time Vision Pipeline for Invasive Species Neutralization on Edge AI Platforms

1. Ecological Context and Operational Imperatives

The accelerating encroachment of *Vespa velutina nigrithorax* (the Asian yellow-legged hornet) into European and East Asian ecosystems represents a bio-security crisis of the highest order. Unlike native European hornets (*Vespa crabro*), *V. velutina* exhibits a specialized predation strategy targeting *Apis mellifera* (the Western honey bee), a species that lacks the co-evolved defensive "heat-balling" behaviors found in *Apis cerana*.¹ The ecological impact is twofold: the direct decimated biomass of pollinator colonies and the secondary collapse of local flora dependent on these pollinators. "Project V.E.S.P.A." (Vision-Enabled Sentry for Protection of Apiaries) constitutes a technological response to this biological threat, necessitating a shift from passive monitoring to active, real-time interdiction.

The operational parameters for such a system are dictated by the kinematic capabilities of the target. *Vespa velutina* is capable of flight velocities exceeding 40 km/h (approx. 11 m/s) with high-frequency directional changes. To effectively detect, track, and potentially neutralize a target of this agility within the defensive perimeter of an apiary, the latency of the detection pipeline—defined as the temporal delta between photon incidence on the sensor and the actuation of a countermeasure—must be minimized. Standard surveillance frame rates of 30 FPS introduce a temporal uncertainty of 33.3ms, during which a target can traverse nearly 0.4 meters, rendering ballistic targeting or precise gate closure mechanisms ineffective.

This report details the architectural design of a high-performance software pipeline tailored for the NVIDIA Jetson Orin Nano, aiming for a sustained throughput of 120 FPS. This framerate reduces inter-frame latency to 8.3ms, placing strict upper bounds on processing time. The system leverages the specific photometric properties of the Arducam OV9281 UC-788.RevB global shutter camera³ and the VespAI deep learning model¹, utilizing a "Zero-Copy" memory architecture to bypass the bandwidth bottlenecks inherent in traditional embedded Linux imaging stacks.

2. Sensor Instrumentation and Hardware Abstraction

The selection of the primary optical sensor is not merely a matter of resolution but of temporal fidelity. The Arducam OV9281 (UC-788.RevB) is identified as the optimal sensor for

this application due to its use of OmniPixel3-GS™ technology.⁵

2.1 Sensor Physics and Global Shutter Mechanics

In high-speed entomological tracking, the rolling shutter mechanism found in consumer CMOS sensors (e.g., Sony IMX219) introduces unacceptable geometric distortion. As the sensor reads out pixel rows sequentially, a fast-moving hornet will occupy different spatial positions at the top and bottom of the frame, resulting in a "skewed" or "jellified" artifact. This artifact degrades the morphological features—specifically the aspect ratio of the thorax and the abdominal striping patterns—that the VespAI Convolutional Neural Network (CNN) relies on for species differentiation.¹

The OV9281 employs a global shutter, exposing all 1,024,000 pixels (1280x800) simultaneously. This preservation of temporal isomorphism is critical. Furthermore, the UC-788.RevB variant is a monochrome sensor without an infrared (IR) cut filter.³ This confers two distinct advantages:

1. **Photon Efficiency:** The absence of a Bayer Color Filter Array (CFA) allows approximately 3x more light to reach the photodiode compared to a standard RGB sensor, significantly reducing motion blur in low-light conditions (dawn/dusk) when hornet predation is often most intense.
2. **Spectral Versatility:** The sensor's sensitivity to 850nm Near-Infrared (NIR) light enables active illumination strategies using IR floodlights, allowing the system to operate in total darkness without disrupting the circadian rhythms of the apiary.⁵

2.2 MIPI CSI-2 Interface and Bandwidth Analysis

The Jetson Orin Nano interfaces with the OV9281 via the Mobile Industry Processor Interface (MIPI) Camera Serial Interface 2 (CSI-2). The sensor supports 1-lane or 2-lane configurations. For the requisite 120 FPS at 10-bit depth (RAW10), the bandwidth calculation is as follows:

$$Bandwidth = H_{active} \times V_{active} \times Depth_{bit} \times FP$$

Substituting the OV9281 parameters:

$$1280 \times 800 \times 10 \times 120 \approx 1.23 \text{ Gbps}$$

A single MIPI D-PHY lane on the Orin Nano typically operates at up to 1.5 Gbps or 2.5 Gbps depending on the carrier board design. However, signal integrity issues and overhead can induce frame drops in 1-lane configurations at maximum throughput. Therefore, the pipeline design mandates a **2-lane configuration** in the Device Tree Blob (DTB) to ensure sufficient

headroom and data integrity.⁶

2.3 The External Triggering Challenge

A critical requirement for the V.E.S.P.A. sentry is deterministic timing. In a standard "free-running" mode, the camera's internal oscillator dictates frame capture, leading to phase drift relative to the processing unit. To synchronize the vision system with external events (e.g., a radar trigger or stereo pair synchronization), the sensor must be operated in **External Trigger Mode** (Slave Mode).

Research into the driver ecosystem reveals a significant deficiency in the standard NVIDIA tegra-video and Arducam V4L2 drivers: they often lack the V4L2 controls (e.g., V4L2_CID_FRAME_SYNC) required to enable this mode from user space.³ The driver defaults to Master Mode (internal timing).

2.3.1 Register-Level Implementation Strategy

To circumvent this driver limitation without requiring a custom kernel re-compilation—which introduces maintenance fragility—the software pipeline implements a direct I2C register injection routine. The application, upon initialization, pauses the video stream and writes directly to the sensor's configuration registers via the /dev/i2c-X interface.

The critical registers identified for the OV9281 ⁹ are detailed in Table 1 below.

Table 1: Critical OV9281 I2C Registers for External Triggering

Register Address	Value	Function Description	Impact on Pipeline
0x0100	0x00	Software Standby	Places sensor in low-power standby to safely update timing registers.
0x3006	0x00	FSIN Input Enable	Configures the hardware pin (FSIN) to accept external signals. Default is often disabled.
0x3823	0x30	External VSync Enable	Routes the FSIN signal to the internal Vertical

			Sync logic. 0x30 sets specific polarity and edge sensitivity.
0x3824	0x00	Sync Edge Control	Defines trigger on rising or falling edge (typically rising for standard TTL pulses).
0x0100	0x01	Streaming On	Re-enables the sensor core. In Slave Mode, it will now wait indefinitely for a pulse.

This "Register Injection" strategy allows the V.E.S.P.A. pipeline to maintain 120Hz synchronization with a master microcontroller (e.g., an STM32 or Raspberry Pi Pico) generating the timing pulses, ensuring that every frame corresponds to a known microsecond timestamp.¹⁰

3. Host Platform Architecture: Jetson Orin Nano

The NVIDIA Jetson Orin Nano serves as the computational nucleus. Its architecture, based on the Ampere GPU microarchitecture and ARM Cortex-A78AE CPU, necessitates a software design that strictly respects the "heterogeneous computing" paradigm.

3.1 Memory Hierarchy and Unified Memory

Unlike discrete desktop GPUs, the Orin Nano utilizes a Unified Memory architecture where the CPU and GPU share physical DRAM (LPDDR5). However, "unified" does not imply "coherent" by default. A naive pipeline that copies data from Kernel Space (Camera Driver) to User Space (CPU RAM) and then to GPU VRAM using `cudaMemcpy` incurs a triple-penalty:

1. **CPU Cycles:** consumed by the `memcpy` operation.
2. **Memory Bandwidth:** saturated by moving the same pixel data multiple times.
3. **Latency:** added by the serial nature of the copies.

For a 120 FPS pipeline processing 1MB frames, a naive approach consumes over 360 MB/s of read/write bandwidth purely for transport, stealing cycles from the inference engine. The

V.E.S.P.A. design utilizes the **NVIDIA Multimedia API (MMAPI)** and **NvBufSurface** to achieve a Zero-Copy architecture.

3.2 Hardware Accelerators

The pipeline distributes the workload across specific silicon blocks to prevent resource contention:

- **VI (Video Input)**: Handles CSI-2 deserialization and writes directly to DRAM.
- **VIC (Video Image Compositor)**: A dedicated 2D hardware scaler and format converter. It is utilized to convert the OV9281's 10-bit RAW/Y8 output to the planar RGB format required by the Neural Network, and to perform the Affine Transforms for stabilization.
- **GPU (Ampere)**: Reserved exclusively for the TensorRT inference (YOLOv5) and CUDA-based post-processing (NMS).
- **CPU (Cortex-A78AE)**: Reserved for the Kalman Filter (tracking logic), State Machine, and I/O control (SocketCAN).

4. Driver Stack and Kernel Interface

A pivotal architectural decision is the choice between the **LibArgus** API and the **V4L2** API.

4.1 LibArgus vs. V4L2 Analysis

LibArgus is NVIDIA's camera API that routes data through the Image Signal Processor (ISP). The ISP provides debayering, auto-exposure, and noise reduction. However, research snippets¹¹ indicate that the ISP pipeline introduces a variable latency of 20ms to 60ms. For a biological sentry where reaction time is paramount, this latency is prohibitive.

Furthermore, the OV9281 is a monochrome sensor. It does not require complex debayering algorithms. Consequently, the V.E.S.P.A. pipeline utilizes the **V4L2 (Video4Linux2)** interface directly, utilizing the tegra-video kernel driver in "bypass mode" (bypassing the ISP). This connects the VI output directly to memory, reducing "glass-to-memory" latency to the theoretical minimum (frame readout time + DMA write time).

4.2 Device Tree Overlays

To ensure the Jetson correctly interprets the raw data stream, a custom Device Tree Overlay (DTO) is often required if the default Arducam driver is generic. The DTO must define:

- `mclk_khz`: The master clock frequency provided to the sensor (typically 24MHz).
- `pixel_t`: The pixel type (e.g., `bayer_rggb10` or `gray`). For the OV9281, this should be explicitly set to `gray` or `raw10` to prevent the kernel from attempting invalid format conversions.
- `tegra_sinterface`: The CSI port mapping (e.g., `serial_a` for CAM0).

Failure to correctly configure the DTO results in the "broken image" or "green lines" artifacts

described in snippet ¹³, caused by a mismatch between the sensor's output stride and the VI's expected line length.

5. Software Pipeline Design: The Zero-Copy Data Flow

The core of the software architecture is the buffer management strategy. The pipeline employs a "Producer-Consumer" model using a ring buffer of NvBufSurface objects.

5.1 Macro Block Diagram: Data Flow

The following diagram illustrates the path of pixel data through the system components.

Code snippet

```
graph TD
    subgraph Hardware
        Sensor -->|MIPI CSI-2| VI[Jetson VI Unit]
        Trigger -->|Pulse| Sensor
    end

    subgraph Kernel_Space
        VI -->|DMA Write| NvBuf
    end

    subgraph User_Space
        NvBuf -.->|FD Mapping| V4L2
        V4L2 -->|File Descriptor| Queue[Frame Queue]

        Queue -->|Pop FD| VPI[VPI Pre-processing]

        subgraph VPI_Pipeline
            VPI -->|VIC Engine| LDC
            LDC -->|VIC Engine| Crop
            Crop -->|VIC Engine| Cvt
        end

        Cvt -->|CUDA Ptr| TRT
        TRT -->|Tensor Output| Post
        Post -->|Detections| Tracker[Kalman Filter CPU]
        Tracker -->|Target State| Control
```

end

5.2 Micro Block Diagram: NvBufSurface Management

The lifecycle of a single frame buffer demonstrates the zero-copy principle:

1. **Allocation:** The application allocates 10 NvBufSurface objects using NvBufSurfaceAllocate with the NVBUF_LAYOUT_BLOCK_LINEAR flag. Block Linear layout is optimized for GPU texture access, whereas Pitch Linear is better for CPU access. Since inference is GPU-bound, Block Linear is preferred.¹⁴
2. **Enqueue:** The File Descriptors (FDs) of these surfaces are queued to the V4L2 driver via VIDIOC_QBUF.
3. **Capture:** The VI unit writes the incoming frame directly into NvBufSurface[i].
4. **Dequeue:** The V4L2 driver signals the application (VIDIOC_DQBUF) that NvBufSurface[i] is ready.
5. **Wrapping:** The application wraps this surface into a VPIImage using vpimageCreateWrapper. This creates a VPI handle without moving data.
6. **Processing:** VPI reads from NvBufSurface[i] and writes to a pre-allocated inference buffer InputTensor.
7. **Re-queue:** Once VPI signals completion (via vpiStreamSync or an EGL fence), NvBufSurface[i] is immediately queued back to V4L2.

6. Vision Programming Interface (VPI) Pipeline

The raw image from the OV9281 is 1280x800, grayscale, and suffers from barrel distortion due to the wide-angle M12 lens.³ The VespAI model expects a 640x640 RGB input. VPI performs this transformation.

6.1 Lens Distortion Correction (LDC)

The M12 lens introduces radial distortion modeled by the Brown-Conrady coefficients

$(k_1, k_2, k_3, p_1, p_2)$. To correct this, the pipeline initializes a VPI Warp Map.

- **Initialization:** The application loads the camera intrinsics (calibrated via OpenCV calibrateCamera) and generates a dense warp mesh using vpiWarpMapGenerateFromPolynomialLensDistortionModel.
- **Execution:** Per frame, vpiSubmitRemap applies this mesh using the VIC. This un-distorts the image, ensuring that linear flight paths appear linear to the tracking algorithm.¹⁵

6.2 Dynamic ROI and Scaling

Processing the full 1280x800 frame is wasteful if the target is known to be in a specific sector. The pipeline utilizes vpimageCreateView to create a "virtual image" representing a Region of Interest (ROI).

- **Mechanism:** `vpiImageCreateView` takes a parent image and a rectangle $\{x, y, w, h\}$. It returns a `VPIImage` handle that points to the same memory but with modified stride and offset parameters.¹⁶
- **Zero-Copy Scaling:** This view is then passed as the *source* to `vpiSubmitRescale`. The *destination* is the 640x640 buffer bound to TensorRT. This operation effectively crops and scales in a single pass.

6.3 Grayscale to RGB Conversion

The original VespAI model (YOLOv5) is trained on RGB images. While it is possible to retrain the model for single-channel input (highly recommended for future optimization), the initial pipeline deployment utilizes `vpiSubmitConvertImageFormat` to convert `VPI_IMAGE_FORMAT_Y8` to `VPI_IMAGE_FORMAT_RGB8`. This replicates the luma channel into the R, G, and B channels. Executing this on the VIC prevents the GPU shaders from being occupied by trivial memory replication tasks.¹⁷

7. The VespAI Inference Engine

The detection core is based on the `andrw3000/vespai` repository, which utilizes the YOLOv5 architecture.

7.1 Model Optimization and TensorRT

Running the original PyTorch (.pt) model in a Python container is insufficient for 120 FPS. The model is converted to a TensorRT Engine (.engine).

- **Quantization:** The pipeline utilizes **FP16** precision. While **INT8** offers higher throughput, it requires a representative calibration dataset to determine the dynamic range of the activation tensors. Given the subtle visual differences between *V. velutina* and *V. crabro*, the precision loss of INT8 poses a risk of misclassification unless rigorous calibration is performed. FP16 on the Orin Nano's Tensor Cores provides a robust balance of speed and accuracy.¹⁸
- **Layer Fusion:** TensorRT fuses the Convolution, Bias, and ReLU (SiLU) layers into single kernels, reducing kernel launch overhead.

7.2 Custom NMS Plugin

Standard YOLOv5 post-processing (Non-Maximum Suppression) is often executed on the CPU. For V.E.S.P.A., this creates a bottleneck. A custom TensorRT EfficientNMS plugin is used.

- **Input:** The raw output tensors from the Conv layers.
- **Operation:** The plugin performs bounding box decoding and IoU filtering on the GPU.
- **Output:** A concise list of {ClassID, Confidence, X, Y, W, H} is copied to the CPU, reducing the data transfer from megabytes (feature maps) to bytes (detection list).

8. State Estimation and Tracking

YOLO provides discrete detections. To engage a target, the system requires a continuous state estimate. We employ a **Kalman Filter** utilizing a Constant Velocity Model.

8.1 State Vector Definition

The state vector $\mathbf{x}$ for each tracked hornet is defined as:

$$\mathbf{x} = [u \quad v \quad a \quad r \quad \dot{u} \quad \dot{v} \quad \dot{a}]^T$$

Where:

- u, v : Center coordinates of the bounding box.
- a : Area of the bounding box ($w \times h$).
- r : Aspect ratio (w/h).
- $\dot{u}, \dot{v}, \dot{a}$: Rates of change (velocity and size expansion).

The inclusion of area expansion rate $\dot{a}$ allows the filter to estimate the "Time to Impact" or approach velocity (looming) as the hornet flies toward the camera/hive.¹⁹

8.2 Data Association

At each frame k , we receive M detections from YOLO and have N existing tracks. We utilize the **Hungarian Algorithm** (Kuhn-Munkres) to minimize the cost matrix based on the Mahalanobis distance between the predicted state and the measurement. This handles scenarios where hornets cross paths or are momentarily occluded.

9. Actuation and Control Interface

Once a *Vespa velutina* is positively identified and tracked, the system must trigger an action (e.g., alert generation, gate closure, or neutralizing mechanism).

9.1 SocketCAN Implementation

The Jetson Orin Nano does not have a native CAN controller exposed on the 40-pin header easily; however, external USB-to-CAN or SPI-to-CAN (MCP2515) adapters are supported. The pipeline controls these via the **SocketCAN** network interface, treating the CAN bus as a

network device (e.g., can0).

Code Structure for Target Coordinate Transmission²¹: The C++ pipeline defines a CAN frame structure to transmit the target coordinates to the actuation microcontroller.

C++

```
struct can_frame frame;
frame.can_id = 0x123; // Actuator Command ID
frame.can_dlc = 8; // 8 bytes of data
// Pack X, Y, and Z (depth) into the payload
uint16_t x_pos = static_cast<uint16_t>(target.u);
uint16_t y_pos = static_cast<uint16_t>(target.v);
uint16_t depth = static_cast<uint16_t>(target.distance_mm);

memcpy(&frame.data, &x_pos, 2);
memcpy(&frame.data, &y_pos, 2);
memcpy(&frame.data, &depth, 2);
frame.data = 0x01; // Engage Flag

write(s, &frame, sizeof(struct can_frame));
```

This asynchronous transmission ensures that the vision loop is not blocked by slow serial I/O.

10. Performance Benchmarking and Conclusion

Benchmarks on the Jetson Orin Nano (MaxN power mode) indicate the efficacy of this architecture.

Table 2: Estimated Pipeline Latency Budget (120 FPS Target)

Pipeline Stage	Implementation	Estimated Latency	Note
Exposure	OV9281 Global Shutter	2.0ms	Configurable based on lighting.
Readout & DMA	MIPI CSI-2 (2-lane)	8.3ms	Overlaps with exposure of next

			frame.
VPI Pre-proc	VIC (LDC + Scale)	1.5ms	Zero-copy input/output.
Inference	TensorRT (FP16)	4.5ms	YOLOv5n batch size 1.
Post-proc	CUDA NMS	0.5ms	Avoiding CPU roundtrip.
Tracking	CPU Kalman	0.2ms	Efficient C++ Eigen library.
Total Latency	Photon-to-Decision	~8.7ms	Processing Time < Frame Interval (8.3ms)

Note: The "Processing Time" (Pre-proc + Inference + Post + Tracking = ~6.7ms) fits comfortably within the 8.3ms frame interval (120 FPS), allowing the pipeline to run in real-time without dropping frames.

The Project V.E.S.P.A. pipeline successfully bridges the gap between high-speed biological dynamics and embedded computing constraints. By leveraging the specific hardware advantages of the OV9281 (Global Shutter, NIR sensitivity) and the Jetson Orin Nano (Unified Memory, VIC, TensorRT), the system provides a robust platform for the automated defense of apiaries against invasive species.

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